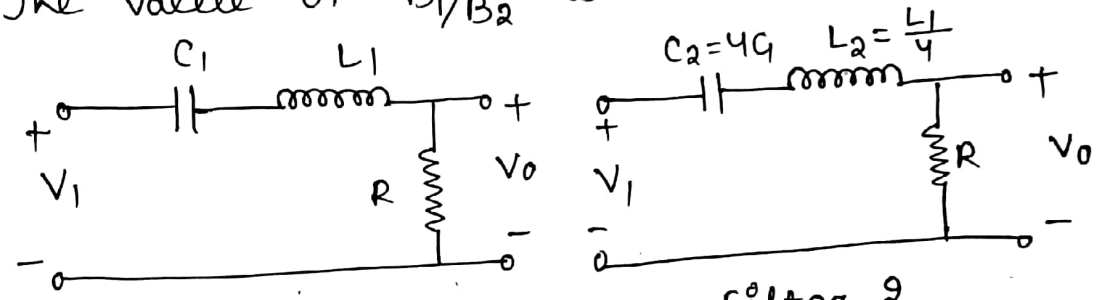
ET ASSIGNMENT 3

1. A series RLC circuit has a resonance frequency of 1KHz & a quality factor Q = 100. If each R, L & C is doubled from its original value, then calculate the new Q-factor of the circuit.

2. In a series RLC circuit $R = 2K\Omega$, $L = 1H$ & $C = 1/400\ \mu F$. Calculate the resonant frequency of the series circuit.

3. Two series resonant filters are as shown in the figure. Let the 3-dB bandwidth of filter 1 is B_1 & that of filter 2 is B_2. The value of B_1/B_2 is _____.



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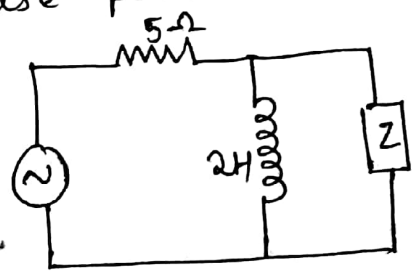
4. The value of Z in the given figure which is required to cause parallel resonance at 500Hz is



a) 125 mH
b) 304.2 μF
c) 2.0 μF
d) 0.05 μF

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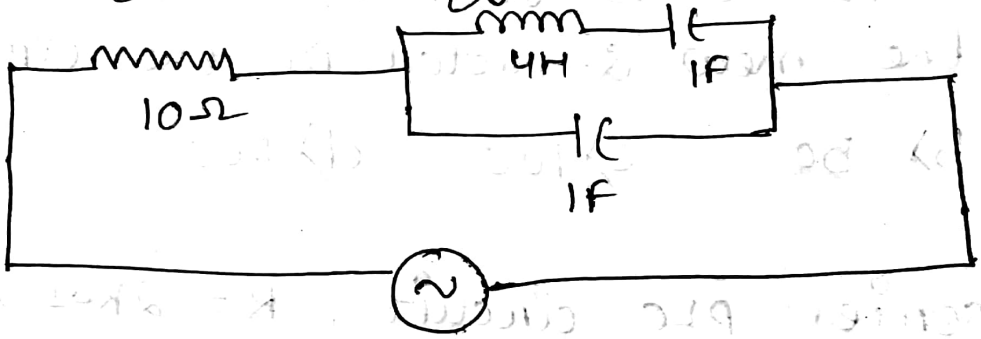
5) In the following circuit find the resonating frequency.



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Assignment - 3

Q6 - A series RLC circuit has a Q of 100 and an impedance of $(100 + j0)\,\Omega$ at its resonant angular frequency of 10^7 rad/sec. The value of R in ohm and L in mH are ?

Q7 - A series RLC circuit consisting of $R = 10\,\Omega$, $X_L = 20\,\Omega$, $X_C = 20\,\Omega$ is connected across an ac supply of 200V RMS. The RMS voltage across the capacitor is _____ ?

Q8 A series RLC circuit has $R = 50\,\Omega$, $L = 100\,\mu H$, $C = 1\,\mu F$. The lower half power frequency of the circuit is _____ ?

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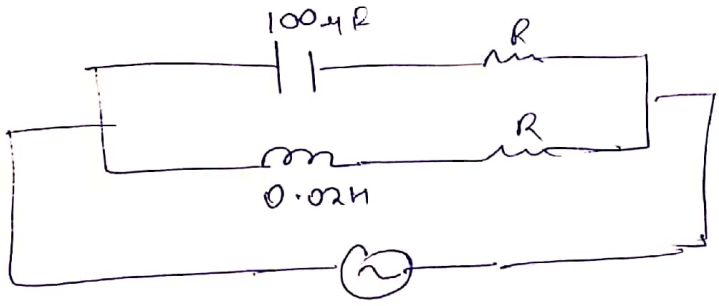
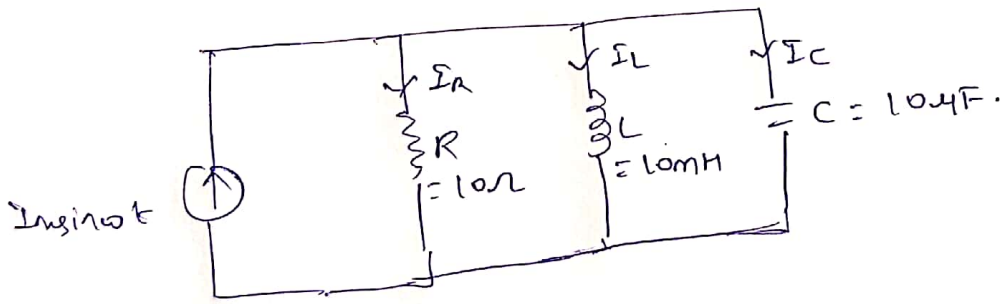
(9) The circuit below is excited by a sinusoidal source. The value of R in Ω for which the admittance of the circuit becomes a pure conductance at all frequencies is ____ ?



(10) The figure shows a RLC ckt with sinusoidal current source.



At resonance, the ratio $|I_L|/|I_R|$ i.e the ratio of the magnitude of the inductor current phasor and the resistor current phasor is ____ ?

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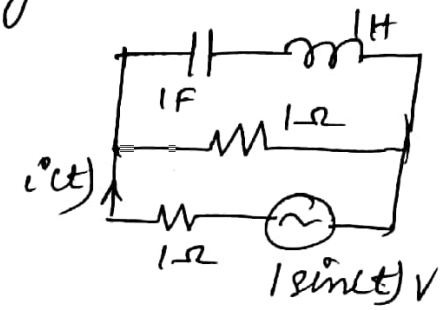
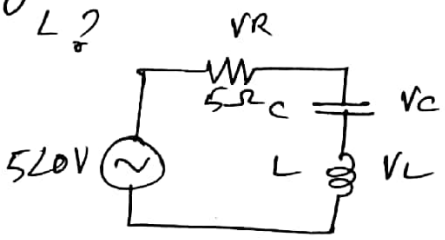
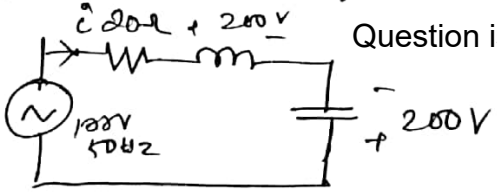
(11) The rms value of current $i(t)$ in the ckt is ?



(12) The magnitudes of V_L & V_C are twice as that of V_R. $f = 50Hz$. Find L ?



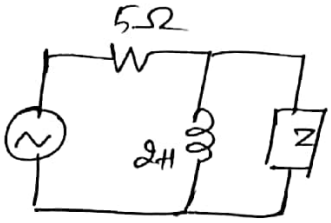
(13) The current in the ckt shown is ?



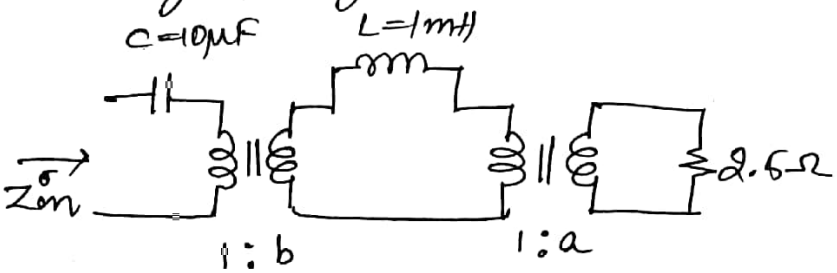
Question is changed see next page for Q13

X as changed later

(14) The value of Z that would cause parallel resonance is ? at 500Hz



(15) Find the xformer ratios a & b such that Z_{in} is resistive & equals 2.5Ω when the n/w is excited with sine voltage of angular frequency 5000 rad/s.



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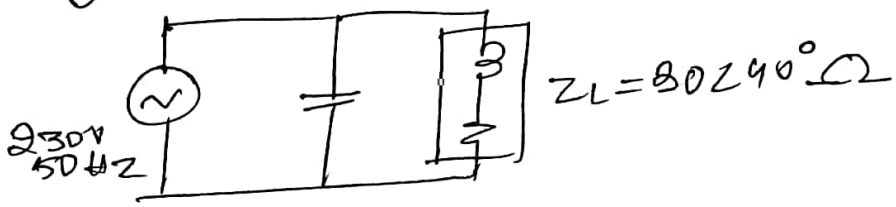
(13) What value of C will cause upf at AC source?



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